Beyond Crest Factor: A Cross-Dataset Causal Audit Protocol for Signal Cues in Speech Anti-Spoofing

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Abstract—Speech anti-spoofing systems can obtain convincing benchmark scores while responding to corpus-specific acoustic cues rather than synthesis artifacts. We present a locked, cross-dataset audit designed to distinguish three different propositions: that a waveform feature separates labels, that it associates with a detector score within label, and that changing it causes a score change under a content-preserving intervention. The study registers 28 interpretable descriptors on three waveform views, eight published detector score artifacts, three discovery corpora, and two held-out confirmation corpora. It tests associations with multiplicity correction and clustered bootstrap intervals, then admits a feature to paired interventions only after a pre-specified portability gate. Conditional on causal evidence, a feature-conditioned, transformation-consistent head is evaluated against an equivalent training recipe. This initial draft reports a first discovery screen in addition to input-integrity results: 71,237 published Spectra-AASIST scores join exactly to ASVspoof 2019 LA labels, and 1,344 within-class associations are screened on a locked balanced subset across eight score artifacts. Crest factor is strongly associated with Spectra-AASIST spoof scores, but its sign and size are heterogeneous across architectures. The evidence therefore motivates the cross-dataset audit; it makes no claim that crest factor, or any other cue, is a portable association or a causal shortcut.

Index Terms—speech deepfake detection, anti-spoofing, shortcut learning, causal audit, robustness

I. Introduction

Speech anti-spoofing countermeasures must distinguish bona fide speech from synthetic, converted, or replayed speech. Modern systems can learn effective spectro-temporal representations; for example, AASIST uses integrated spectro-temporal graph attention [1]. However, strong scores do not by themselves establish that a system responds to the intended manipulation artifacts. Müller et al. found that silence duration can correlate with the target label in ASVspoof data and that this cue alone can yield strong discrimination [2]. More generally, controlled shortcut conditions can produce near-perfect or worse-than-chance countermeasures without a reliable signal of the intended task competence [3].

The present study begins with a practical observation: an interpretable feature—including crest factor—may be label-discriminative, correlated with a detector score, and causally influential under waveform transformation; these are different claims. Confusing them risks attributing model reliance to a feature that merely tracks label prevalence, corpus identity, duration, or loudness. We therefore audit published per-sample scores without recomputing baseline inference and use a staged, locked decision process.

This paper makes the following methodological contributions.

- It registers a cross-dataset, multi-model association design that separates label effects from within-class score-feature associations and fixes the acceptance criterion before confirmation data are analyzed.
- It specifies quality-gated paired waveform interventions that test causal score sensitivity rather than treating correlation as a shortcut.
- It defines a conditional mitigation experiment: a feature-conditioned head with transformation consistency is evaluated only as a consequence of audit evidence, against an equivalent training recipe and held-out corpora.

The manuscript is intentionally an initial, evidence-conservative draft. Sections III–IV distinguish locked methodology, the first discovery screen, and pending confirmation. The title and narrative will be updated only after the documented outer-loop evidence gate.

II. Background and audit target

The ASVspoof 2019 challenge provides a common setting for logical-access spoofing research [4]. Let $y_i \in \{0, 1\}$ denote the registered label of utterance i, f_i an interpretable waveform feature, and $s_{m,i}$ the published score of model m. Each score artifact is oriented to increasing spoof evidence only after a label-direction check; artifacts that fail an exact sample-ID join or an orientation check are excluded rather than heuristically repaired. Let c_i contain registered nuisance variables such as duration, integrated loudness, and available speaker, attack, or codec metadata.

The first audit target is the within-class association $\rho_{m,y} = \text{Spearman}(f_i, s_{m,i} \mid y_i = y)$ and its rank-residual analogue after controlling for c_i. This target is not a causal estimand: a residual association may still be caused by unobserved corpus properties. A subsequent paired transformation supplies the causal target $\Delta s_{m,i} = s_m(T(f)_i) - s_m(x_i)$, subject to quality gates that limit changes to speech content and loudness. The distinction structures the three hypotheses below.

III. Locked study design

A. Inputs, feature registry, and provenance

All datasets, model revisions, and published score artifacts are pinned in the repository manifest. Discovery uses ASVspoof 2019 LA, ASVspoof 2021 LA, and ASVspoof 2021 DF; In-the-Wild and ASVspoof 5 are held out for confirmation. The score panel consists of Spectra-AASIST, AASIST, Res2TCNGuard, RawTFNet, WhisperMFCCMesoNet, W2V2-AASIST, XLSR-SLS, and RawBMamba. Where a published score artifact is absent, the cell is reported as missing and is never imputed. This design uses the Arena score artifacts as published evidence, rather than spending additional compute to reproduce their baseline inference.

The frozen v1_28 registry extracts 28 amplitude, spectral, excitation, phase, and modulation descriptors from full-waveform, deterministic-crop, and pre-emphasized views. Voice-dependent descriptors retain missing values for unvoiced or unsuitable signals; they are not converted to zero. For corpora above 50,000 examples, the analysis selects a deterministic, class-balanced subset using seed 2609 and at most 5,000 examples per class.

B. H1: association and portability

For each model, dataset, class, feature, and waveform view, H1 reports feature label AUROC, within-class Spearman association, and partial rank association. P-values are adjusted with Benjamini–Hochberg correction. Candidate estimates receive 2,000 clustered bootstrap replicates, with a speaker or source utterance cluster when metadata permits. Exclusions are limited to missing stable IDs, unreadable waveforms, required labels, or published scores; every count is reported.

A feature is portable only if it has the same direction in at least four of the five core datasets, is significant in both held-out confirmation datasets, and is present in at least five of eight models. This criterion is fixed before confirmation analysis. Crest factor is one candidate in this registry, not the pre-supposed conclusion.

C. H2: quality-gated paired interventions

H2 is run only on H1-frozen candidates, with 500 examples per class per core dataset and the intervention panel Spectra-AASIST, AASIST, Res2TCNGuard, and W2V2-AASIST. The registered transformations comprise loudness-matched dynamic-range compression, leading/trailing silence standardization, spectral tilt, stable all-pass phase perturbation, and gain and polarity negative controls. A transformed example is retained only if STOI is at least 0.95, original-to-transformed transcript WER is at most 5%, loudness drift is at most 0.2 LU, and clipping is absent. An arm is dropped if fewer than 90% of its examples pass. Causal sensitivity requires a nonzero bootstrap interval, a standardized paired score change of at least 0.2, and consistent direction in at least three datasets and three of four models.

TABLE I
Completed evidence at the initial-draft checkpoint. The H1 row is a discovery observation; it is not a portability or causal claim.

Artifact	Observed status
Spectra-AASIST score/label join on pinned ASVspoof 2019 LA	71,237 scores join one-to-one; raw score is oriented toward bona fide evidence and is negated for the canonical spoof-evidence analysis.
Feature-registry pilot on deterministic ASVspoof 2019 LA sample	200 utterances $\times$ 3 waveform views; non-voice descriptors complete and voice-quality descriptors available for 95% of records.
H1 discovery screen on locked ASVspoof 2019 LA subset	1,344 within-class tests (8 models $\times$ 2 classes $\times$ 3 views $\times$ 28 features). For spoof trials, Spectra-AASIST crest-factor partial ρ is -0.454, -0.445, and -0.327 by view; signs and magnitudes differ across the panel.
Cross-dataset H1 portability, H2 paired deltas, and H3 held-out EER	Pending; no portability, causal, or improvement conclusion is claimed.

D. H3: conditional feature-conditioned mitigation

H3 is confirmatory only after an H2 causal result; otherwise it is explicitly exploratory. Using ASVspoof 2019 LA train/dev only for training and tuning, we compare an equivalent-recipe baseline head, logistic late score-feature fusion, a learned feature-conditioned head, and that head with consistency loss on H2-accepted transformations. Spectra-AASIST and AASIST are the first backbones. Each newly trained variant uses BF16, seeds 13/37/73, and an independently measured fastest batch/worker setting on one RTX 6000 Ada GPU. Success requires at least 10% relative macro held-out EER improvement on two or more datasets, a paired interval excluding zero, and no more than a 0.2-point in-domain EER degradation.

IV. Completed integrity checks and pending results

Table I records the completed empirical artifacts at this initial-draft checkpoint. The first H1 row is a discovery screen, not a cross-dataset conclusion: it is included because it has a compact result artifact and analysis note. No intervention effect, EER, or model ranking is reported before its corresponding artifact exists.

V. Limitations and responsible reporting

The proposed audit cannot prove that every unmeasured cue is irrelevant, and its interventions cover only registered transformations that pass automated quality gates. Metadata availability differs by corpus, so the

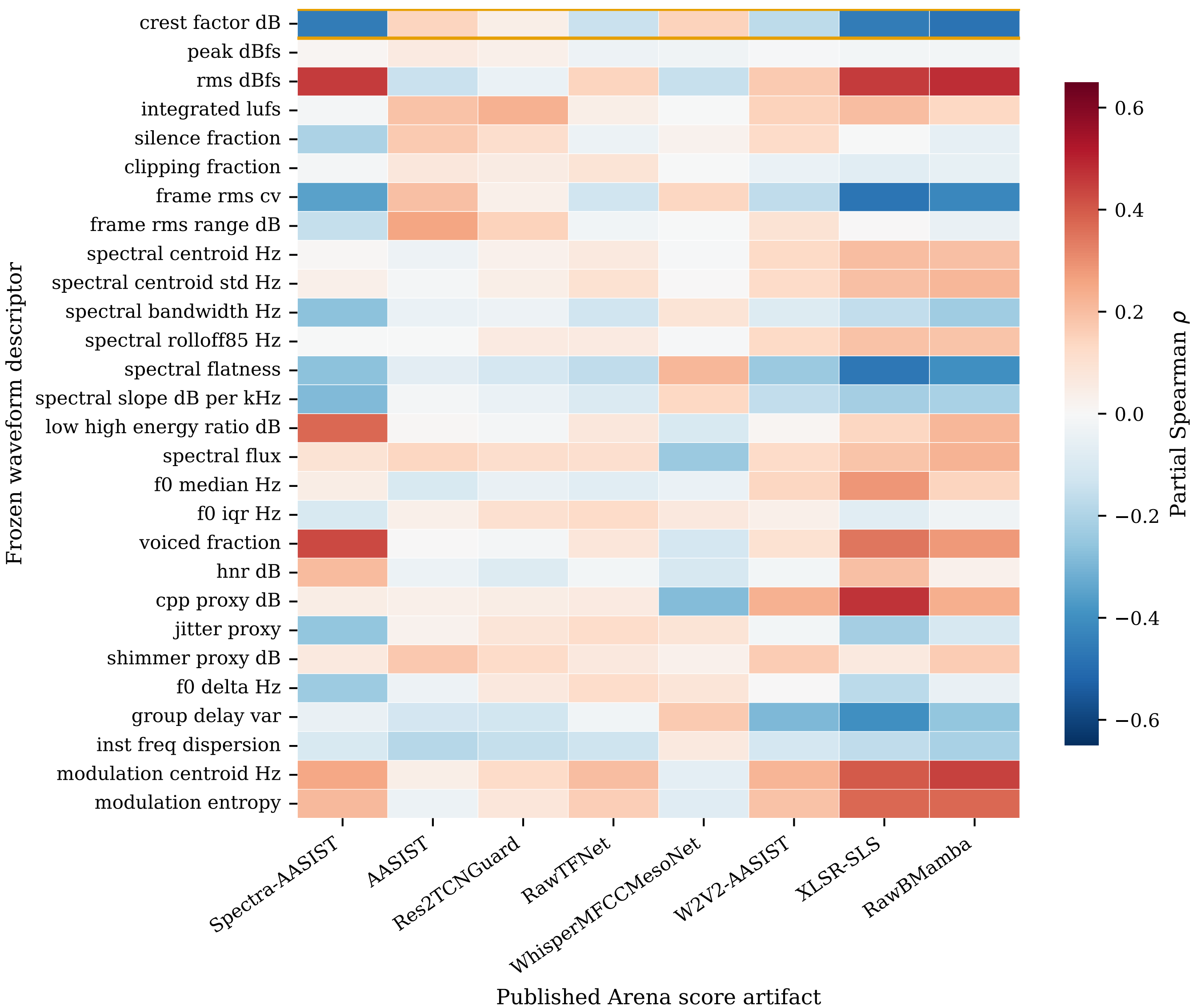

Fig. 1. Complete first-discovery association screen for spoof trials on the locked ASVspoof2019 LA subset. Each cell is a duration-, loudness-, and metadata-adjusted partial Spearman association between a frozen waveform descriptor and the published detector's canonical spoof-evidence score. All 28 descriptors and all eight available score artifacts are shown; the orange outline marks crest factor. The heterogeneity across columns is the result: this one-corpus discovery plot does not support a universal cue-reliance claim.

partial analyses must report which controls are present rather than implying uniform adjustment. Published score artifacts broaden architecture coverage but do not replace access to all model internals. The conditional mitigation can establish only a bounded robustness result under the registered splits and transformations, not universal safety against new generators or post-processing pipelines.

The current artifact set also has a more immediate limitation: the sole H1 screen comes from one discovery corpus. Its extremely small adjusted p-values reflect 5,000 trials per within-class test and cannot substitute for effect-size, cross-corpus, and held-out evidence. This draft therefore treats the observed crest-factor association as a heterogeneous discovery pattern, not a conclusion about detector reliance. The final paper will either report the full held-out results under this protocol or clearly present a negative result and its scope.

VI. Conclusion

We present a reproducible path from an observed score-feature correlation to a testable claim about detector sensitivity: exact artifact joins, within-class and adjusted associations, quality-gated paired interventions, and a conditional robustness test. The first discovery screen shows that crest-factor associations can be strong yet architecture-specific, illustrating why a single-model correlation cannot establish shortcut reliance. The next evidence-bearing milestone is the locked H1 matrix across the remaining discovery and held-out corpora; until then, the paper makes no portability or causal feature-reliance claim.

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